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E-222 UNINTERRUPTIBLE POWER SUPPLY

1. SCOPE OF WORK

The Work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, testing and commissioning of the uninterruptible power supplies in accordance with Part 4, Subsection 1.2 Summary of Works.

Complete set of power supplies, for gas turbine unit and for steam turbine units as shown in the drawing, with single phase, 230 V, 50 Hz, 0.8 lagging to 1.0 PF, including, but not be limited to; the associated input rectifiers, auctioneering diodes, inverters, alternate supply voltage regulating transformers, static transfer switches and the output power distribution panels.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards” and the following specific standards:

2.1 Institute of Electrical and Electronics Engineers

IEEE 587 : Guide for surge voltages in low-voltage AC
Power circuits.

2.2 International Electrotechnical Commission

IEC 60044 : Instrument transformers
IEC 60076 : Transformers
IEC 60146 : Semiconductor rectifiers and convertors
IEC 60529 : Classification of degrees of protection provided by
enclosures (IP Code)
IEC 60726 : Dry type power transformers
IEC 60947 : Low-voltage switchgear and controlgear

3. OPERATING AND DESIGN CONDITIONS

3.1 General

The station UPS shall furnish highly reliable 230 V AC auxiliary power for vital electrical loads in the power plant.

The uninterruptible power system shall operate on unit basis, with the adequate capacity *) kVA, UPSs shall be furnished for the steam turbines power plant units. It is intended to have the UPS loads of each generating unit in two groups to minimize the possibility of adverse interaction between loads having switching type internal power supplies. (*) Item to be determined by the Contractor and subject to the Owner's approval.

Each UPS shall be an integrated system consisting of two (2) 230 V static inverter with a static switch, two (2) rectifiers, two (2) diode auctioneering inputs, circuit breakers, one (1) alternate supply transformer and line voltage regulator, a manual bypass switch for AC output, and a fused distribution panel.

3.2 Power Feeds

The Contractor shall provide two (2) sets of separate power source of 400 V, three-phase, 50 Hz power feeds to UPS through rectifiers. This three-phase feeder shall be the normal source of power.

The Contractor shall also provide one (1) 220 V DC power feeder from the 220 V DC ungrounded station battery that will serve as the power supply to each inverter when the normal AC power to the rectifier is interrupted.

The 220 V DC ungrounded station battery will serve a variety of loads in addition to the inverter and will be subject to a variety of disturbances. These disturbances may include a steady-state voltage variation from 178 to 242 V, 100% voltage dips of up to twenty (20) milliseconds due to short circuits on the DC distribution system and voltage spikes up to 4000 V for ten (10) microseconds on a 40 Ω impedance.

4. DETAILED REQUIREMENTS

Uninterruptible Power Supply (UPS) shall be manufacture or assembled Indonesia.

4.1 Rating

The output rating of each UPS shall be adequate capacity *) kVA, single-phase, 230 V, two-wire, 50 Hz, 0.8 lagging PF to 1.0 PF, at 40°C ambient. (*) Item to be determined by the Contractor and subject to the Owner's approval).

The equipment shall be capable of carrying overloads, at the rated output voltage, of up to 125% for two (2) hours with an ambient temperature not exceeding 30°C and to 150% for one (1) cycle (to provide adequate sub cycle fault current for fuse clearing on short circuits) when supplied from the AC or DC source at an ambient temperature not exceeding 40°C. For overloads in excess of approximately 125% and 150% in excess of one (1) cycle, the UPS system shall be electronically current limited.

All components shall be matched to the specified rating and the equipment shall be of the heavy duty type for central power station service.

4.2 Inverters

The inverter voltage regulation (transient response) shall not exceed the following limits under the range of conditions specified in Clause 3 herein, with loads of 0.8 lagging to 1.0 PF (there will be no leading power factor loads) and other variables as stipulated herein:

- (1) For steady state loads $\pm 2.0\%$ from 0 to 100% full rated load, 20 to 40°C ambient, 178 to 242 V DC input.
- (2) For sudden application or removal of 100 percent of full rated load, the change in inverter output voltage shall not exceed the following:
 - a. $\pm 10\%$ after 0.5 cycle.
 - b. $\pm 2.5\%$ after 1 cycle

The clearing of faults on the branch circuit protected by the largest fuse shall result in a maximum average voltage deviation not exceeding $\pm 10\%$ after 0.5 cycle following initiation of a fault.

The output voltage shall be adjustable over a range of $\pm 5\%$ from the normal voltage or two, 2.5 % taps, above and below the normal turns ratio, are to be supplied for fixed output type inverters.

The output waveform of the inverter shall have less than 5% distortion from a pure sine wave at the worst case combination of DC input voltage variation from 178 to 242 V, 10 to 100% load, and 0.8 lagging to 1.0 PF, excluding a fault condition.

The inverter shall be equipped with a 50 Hz $\pm$ 0.5% internal frequency control. The maximum slew rate shall not exceed 2.0 Hz per second.

The inverter shall be equipped with automatic synchronizing controls. The inverter shall normally operate in synchronism with the alternate AC power supply. Synchronism shall cease when the frequency of the alternate AC power supply changes $\pm$ 1% or upon its loss. The inverter shall then operate on its internal frequency control. Upon return of normal frequency and voltage, the inverter output voltage shall pull back into synchronism with AC supply at the specified slew rate after a time delay of approximately 1.0 second.

The efficiency of the inverter shall be in excess of 80% at full rated load and unity power factor.

An auctioneering diode arrangement shall be furnished and arranged so that the rectifier will not charge the station battery. The DC supplies will remain independent of each other at all times, system disturbances such as grounds, spikes and short circuits on one supply will not be transmitted to the other. The auctioneering diodes shall prevent the input capacitors from discharging into a fault on either DC system.

4.3 Rectifier Unit

The rectifier shall be used to supply normal DC power to the static inverter.

The rectifier shall be voltage regulated, full wave silicon type. The rectifier shall be capable of carrying the continuous, short time (over load) and short circuit currents specified for the UPS system. The rectifier output voltage shall be adjustable over a range of 200 to 250 V.

The voltage regulation shall be less than $\pm$ 1% from no load to full load with a $\pm$ 10% variation in supply voltage. The rectifier shall function as specified with a $\pm$ 3% variation in supply voltage frequency.

The rectifier shall be isolated from the AC power systems by an isolating transformer integrated into the UPS design.

4.4 Static Transfer Switch

The static transfer switch shall be capable of carrying the continuous, short time (overload) and short circuit currents of the UPS.

The switch shall be used for automatic transfer between operating inverter and standby inverter. The switching shall be done with silicon controlled rectifiers or other agreed upon solid-state circuitry.

The switch shall have an independent and continuously adjustable range for automatic under voltage transfer and automatic return on normal voltage. The initial factory setting shall be such that transfer to the alternate source occurs at $\pm 10\%$ of the nominal voltage. After the normal source voltage (at $- 5\%$ of the nominal voltage) is established, the load shall automatically transfer to the normal source. The nominal voltage shall be 230 V AC.

An approximate 2.0 seconds time delay shall be provided before automatic transfer to the normal or inverter source of power after an inverter failure has occurred. During this 2.0 seconds period, to assure source stability, the normal or inverter source voltage shall remain within $\pm 5\%$ of the nominal voltage and $\pm 0.8\%$ of the nominal frequency; also the source must be synchronized at time of transfer.

A pilot relay shall be provided for remote indication of the operational status of the static switch. The relay shall have two double-poles, double-throw contacts.

A static switch continuity monitor and latch circuit to prevent the static switch from returning to the normal supply after an internal fault had developed shall be included. The continuity monitor shall monitor the normal supply constantly. If an internal fault at the normal supply is detected by the continuity monitor, the latch circuit shall transfer and latch the static switch to the alternate supply. A manual reset (pushbutton or equal) shall be included to manually return the static switch to the normal supply after an internal fault is removed. For faults external to the static switch, return shall be automatic.

4.5 AC Distribution Panel

Distribution panel 230 V, single phase, 50 Hz, two-wire, with 200 ampere main and neutral buses and the following single-pole branch circuits:

36 - 30 A distribution single pole branch switches

The panel shall be furnished with a hinged single door with a lock and key, card holders for circuit identification and trim over the entire panel.

The distribution panel shall be factory wired to other related equipment. The buses and interconnections shall be suitable for carrying full inverter output without exceeding 65°C temperature rise.

The circuits and arrangement of the distribution panel shall be subject to the Owner's review and acceptance.

4.6 Miscellaneous Requirements

The devices shall include but not be limited to the following:

- (1) Provisions shall be available for testing operation of inverter while in operation without inadvertently cutting off power to the load.
- (2) Circuit breakers for each pole of each DC and AC supply.
- (3) Manual bypass switch shall be provided for cutting out and isolating the various components of the system without interrupting service to the load.
- (4) Adjustments (potentiometer or taps) for output voltage from inverter and voltage regulator.
- (5) Adjustments for the voltage at which transfers are initiated, forward (to alternate supply) and reverse (to inverter).
- (6) Instruments of Accuracy Class 2 and of appropriate ranges shall be furnished for the following services:
 - a. Voltmeters:
 - One (1) for each battery input to equipment, suitably fused.
 - One (1) for rectifier output, suitably fused.
 - One (1) for alternate source input, suitably fused.
 - One (1) for distribution panel input, suitably fused.

- b. Ammeters:
 - One (1) for each battery input to equipment with suitable shunt.
 - One (1) for rectifier output with suitable shunt.
 - One (1) for distribution panel input with suitable current transformer.
 - c. Frequency meter:
 - One (1) for inverter output suitably fused.
- (7) The following alarm contacts shall be furnished for use with remote annunciator:
- a. Low DC input voltage.
 - b. Low AC input voltage.
 - c. Transfer from inverter to alternate supply.
 - d. Inverter failure.
 - e. High ambient air temperature inside the cabinet.
 - f. Failure of a cooling system (if applicable).
 - g. Rectifier failure.
 - h. Other alarms at the option of the manufacturers.
- (8) Instruments

Instrument potential coils shall be 300 V, 50 Hz for AC and 300 V DC, and AC current coils shall be 5 ampere, 50 Hz.

Current transformers shall have accuracies in accordance with IEC standards. The primary and secondary of any voltage transformers shall be fused.

All instrument transformers shall grounded directly to the ground bus.

The DC ammeters shall have 50 or 100 millivolt shunts and the necessary shunt leads.

- (9) Pilot lights shall be furnished on the front of cabinet to indicate:
 - a. Normal source supplying load
 - b. Alternate source supplying load.
 - c. Continuity failure.

- (10) Provisions shall be furnished on front of cabinet to manually transfer the static switch from the normal to the alternate source or vice versa.

5. PERFORMANCE TESTING AND GUARANTEES

The completely assembled equipment shall be given a simulated operational test at the factory in order to ensure compliance with the operational requirements of this specification.

The input and output insulation levels shall be capable of withstanding a test potential of 1000 volts plus two times the line voltage between current-carrying and non-current carrying parts for one minute on the power circuits.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control shall conform to the requirements in Part 4, Subsection 2.7.

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